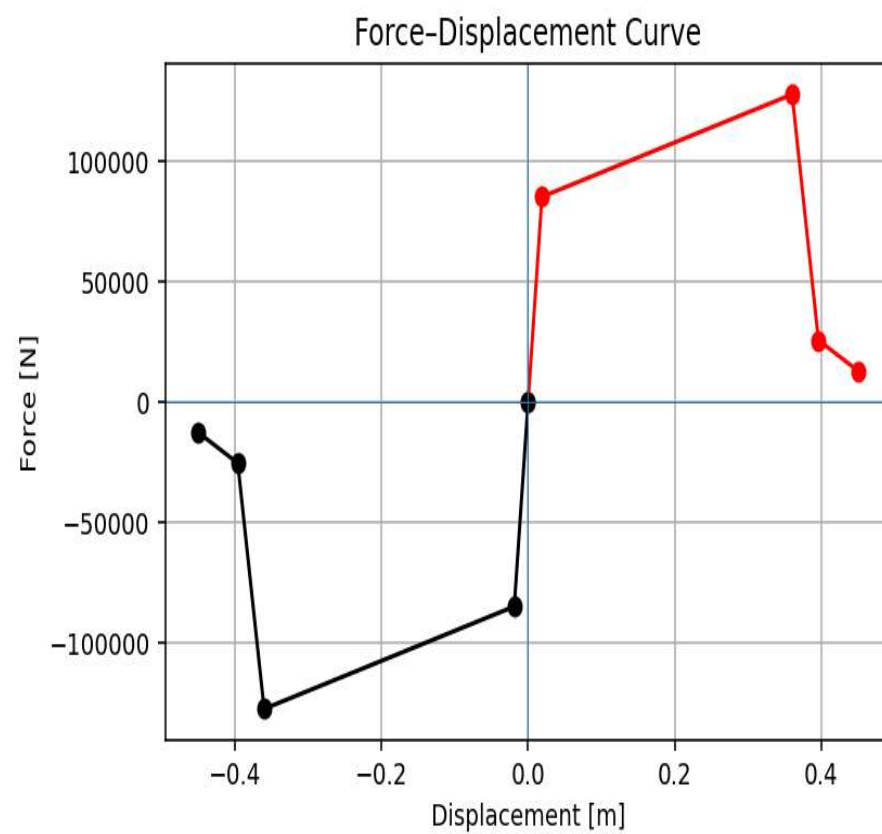
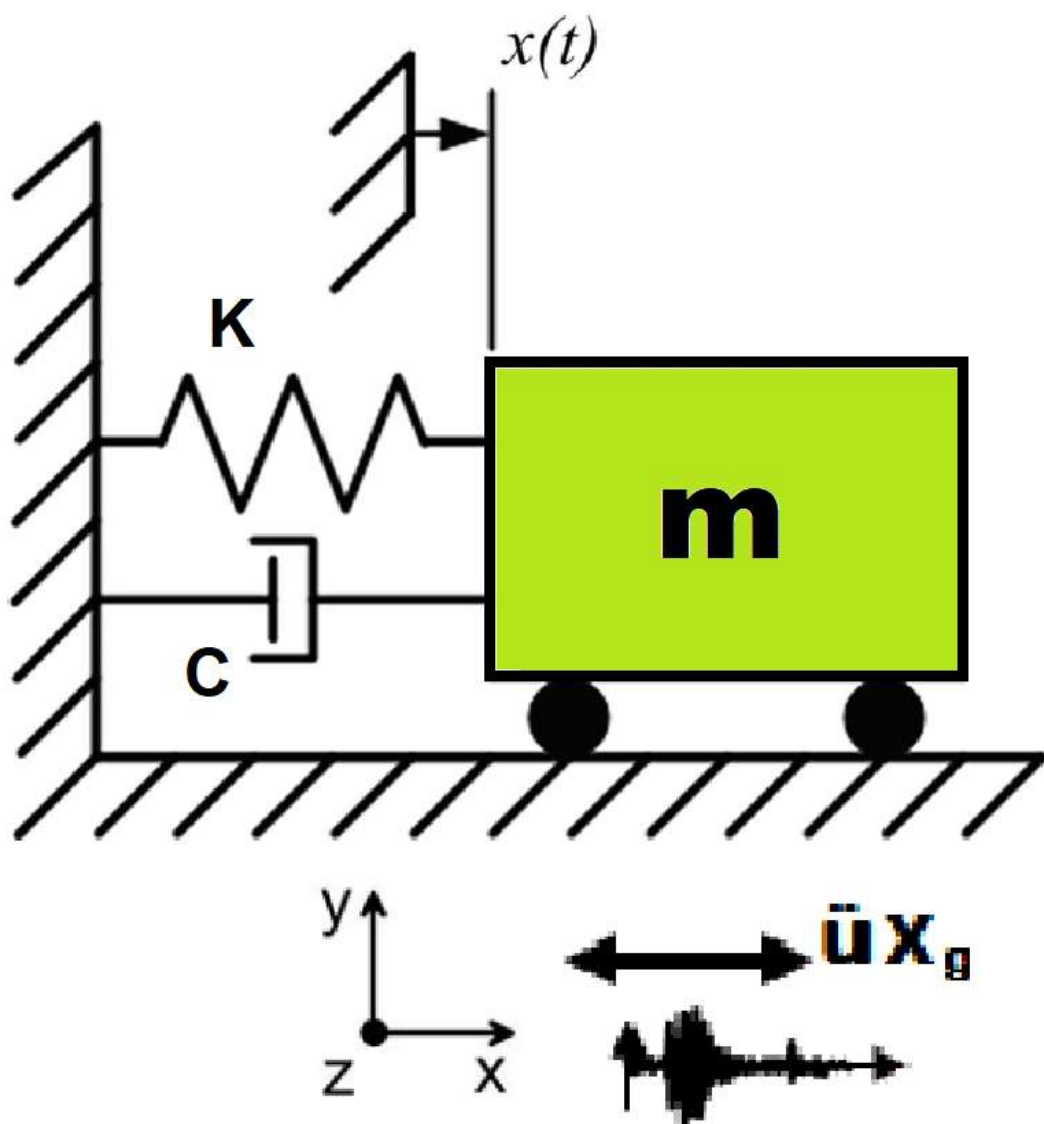


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

**FRAGILITY ANALYSIS BASED ON
ACCELERATION, STRUCTURAL DUCTILITY
DAMAGE INDEX, ENERGY DISSIPATION
CAPACITY INDEX, EQUIVALENT VISCOUS
DAMPING RATIO WITH NONLINEAR DYNAMIC
ANALYSIS OF A SINGLE-DEGREE-OF-FREEDOM
(SDOF) SYSTEM UTILIZING 30 GROUND
MOTIONS IN OPENSEES**

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

Δ_u = Lateral Ultimate Displacement from Pushover Analysis

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+HEIGHT_ANALY...DCI_EVDR\SDOF_SDOF_RESPONSE_SPECTRUM_EDCI_EVDR.py

SDOF_SDOF_RESPONSE_SPECTRUM_EDCI_EVDR.py x EQUIVALENT_VISCOUS_DAMPING_RATIO_FUN.py x

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # FRAGILITY ANALYSIS BASED ON ACCELERATION, STRUCTURAL DUCTILITY DAMAGE INDEX, ENERGY DIS
4 # CAPACITY INDEX, EQUIVALENT VISCOUS DAMPING RATIO WITH NONLINEAR DYNAMIC ANALYSIS (
5 # OF A SINGLE-DEGREE-OF-FREEDOM (SDOF) SYSTEM UTILIZING 30 GROUND MOTIONS IN OPENSEE
6 # -----
7 # INELASTIC RESPONSE SPECTRUM ANALYSIS
8 # -----
9 # EQUIVALENT VISCOUS DAMPING RATIO:  $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$ 
10 # -----
11 # ENERGY DISSIPATION CAPACITY INDEX =  $100 * E_d(\text{earthquake}) / E_d(\text{cyclic displacemen}$ 
12 # -----
13 # This program performs Dynamic Analysis on a Single-Degree-of-Freedom (SDOF) system
14 # subjected to 30 seismic ground motions. The analysis evaluates the structural response unde
15 # levels of seismic intensity.
16 # The framework is designed to support researchers and engineers in assessing the probabilist
17 # performance of structures, with a focus on understanding the impact of uncertainty on struc
18 # response and design.
19 # -----
20 # Key Features:
21 # - Simulation of SDOF system using OpenSees.
22 # - Incremental scaling of ground motions for Nonlinear Dynamic Analysis.
23 # - Probabilistic fragility assessment based on predefined damage states.
24 # - Visualization of structural response and fragility curves.
25 # - Export of results for further analysis.
26 # -----
27 # THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
28 # EMAIL: salar.d.ghashghaei@gmail.com
29 #####
30
31 """
32 This code implements a comprehensive nonlinear dynamic analysis framework for
33 performance-based earthquake engineering assessment of single-degree-of-freedom
```

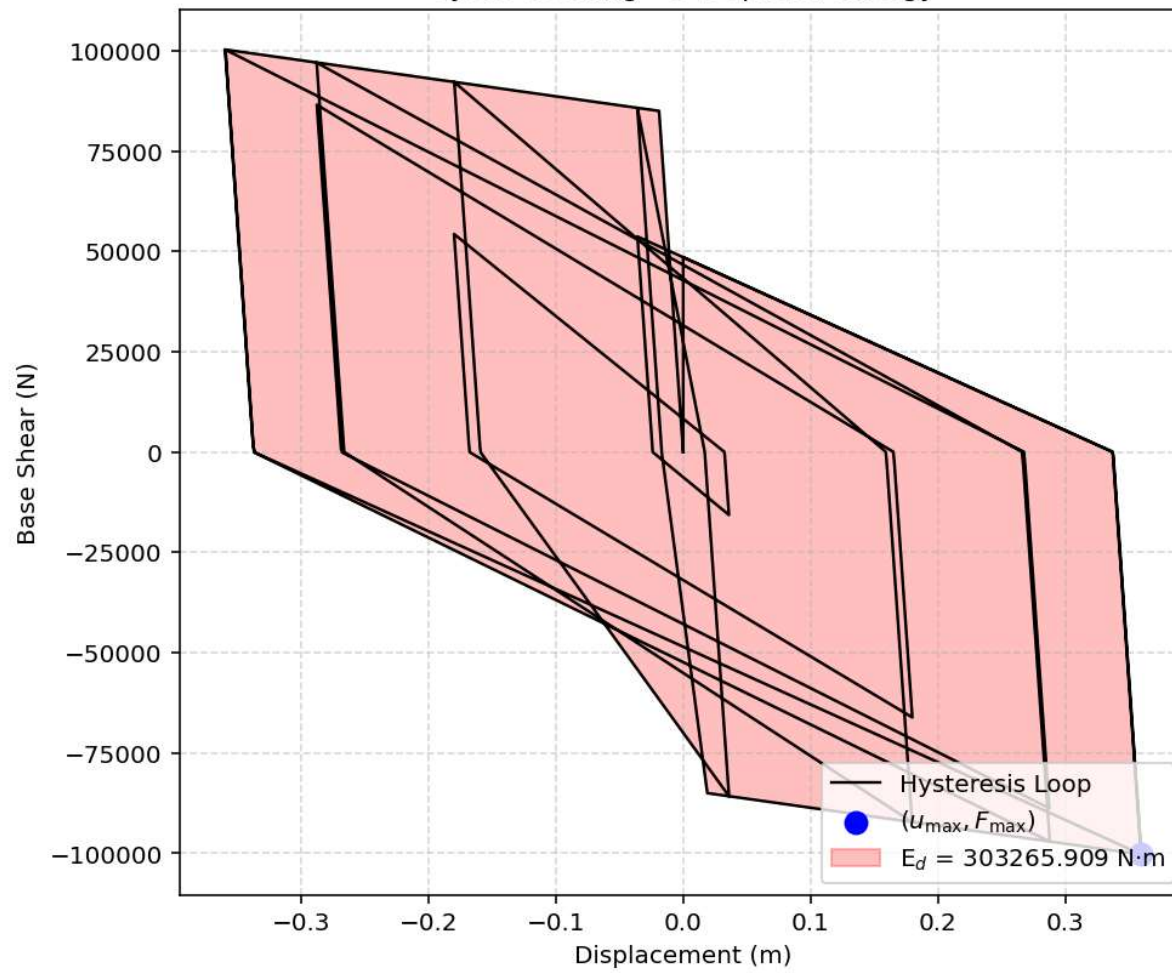
Options

Displacement Response Spectrum

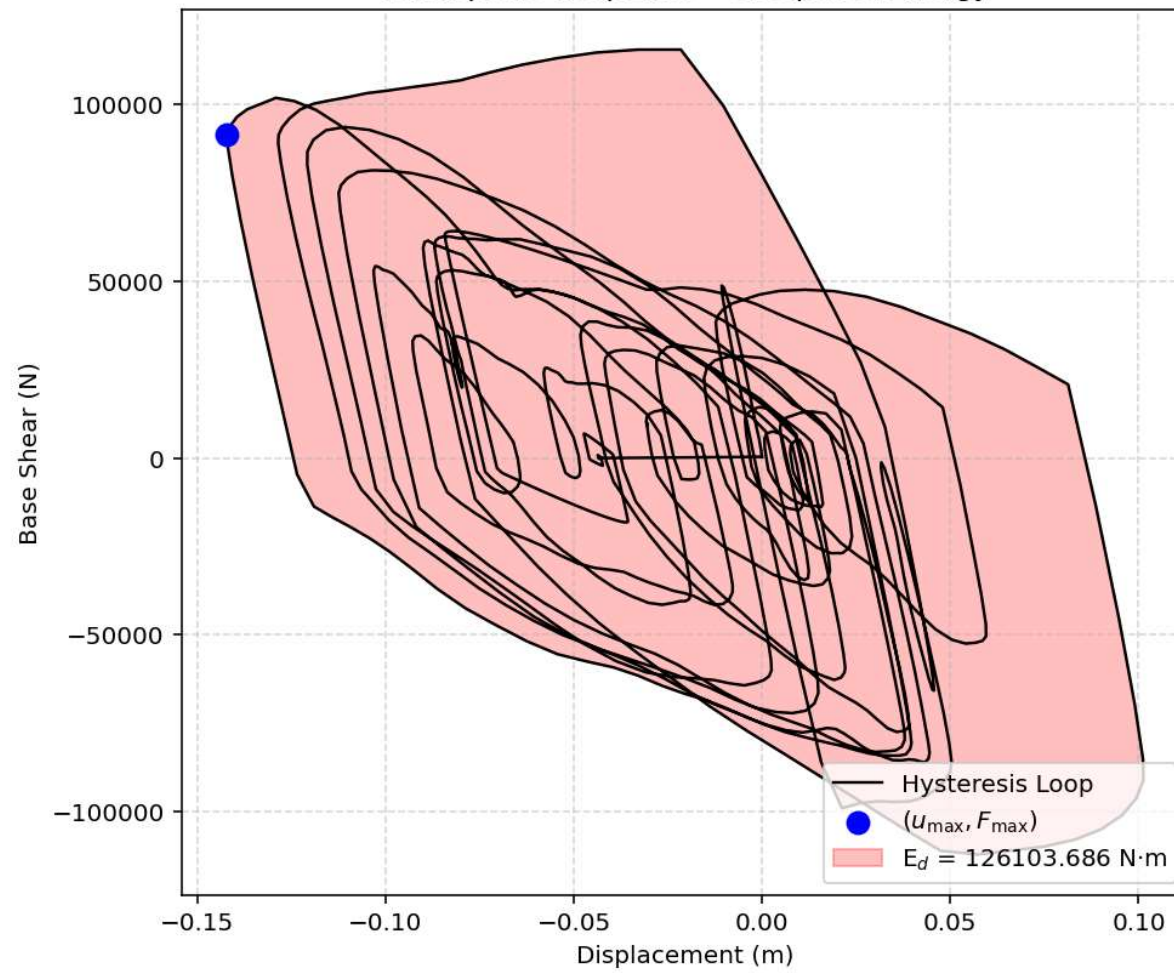
IPython Console Plots Variable Explorer Help Debugger History Files

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 156, Col 16 UTF-8 CRLF RW Mem 66%

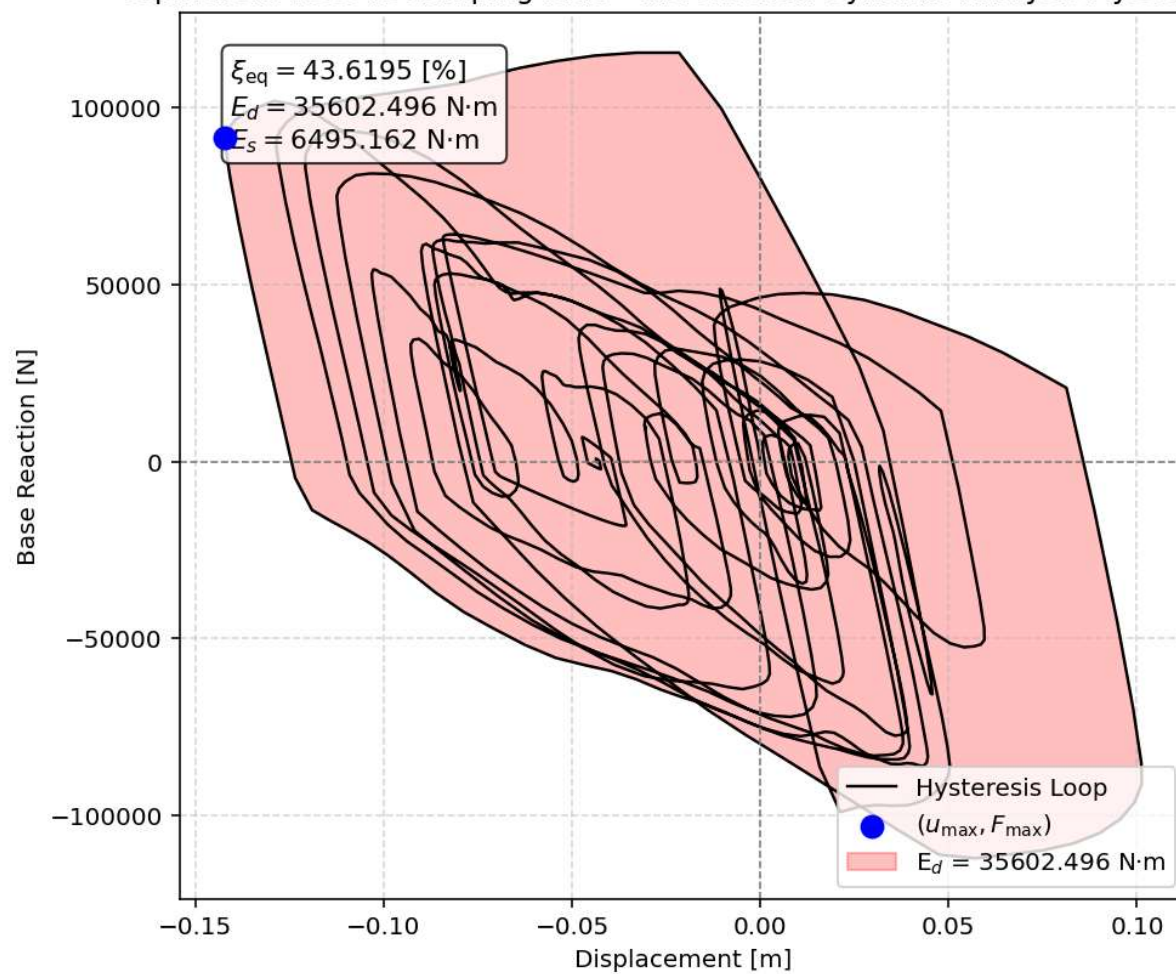
Cyclic Loading - Dissipated Energy

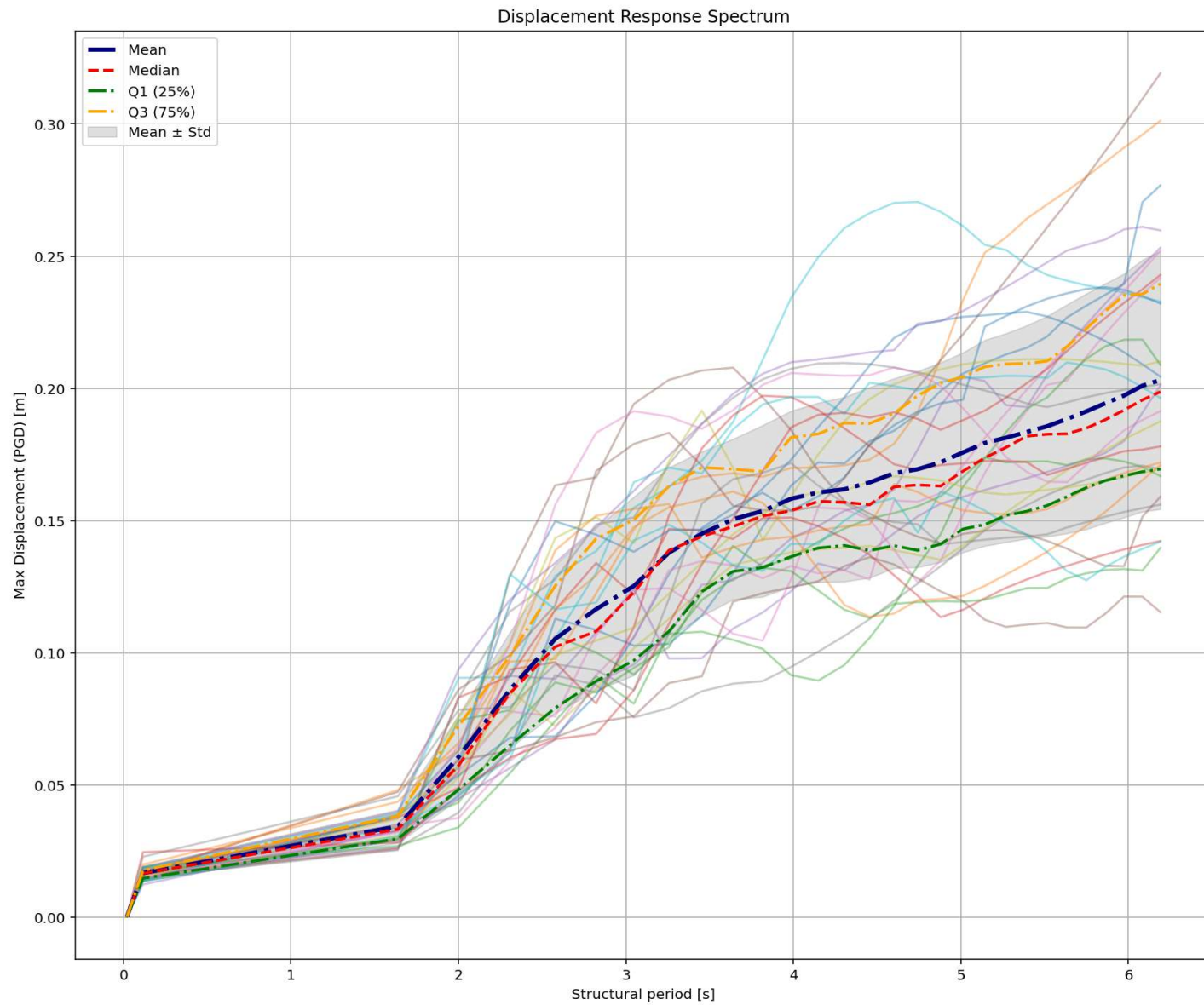


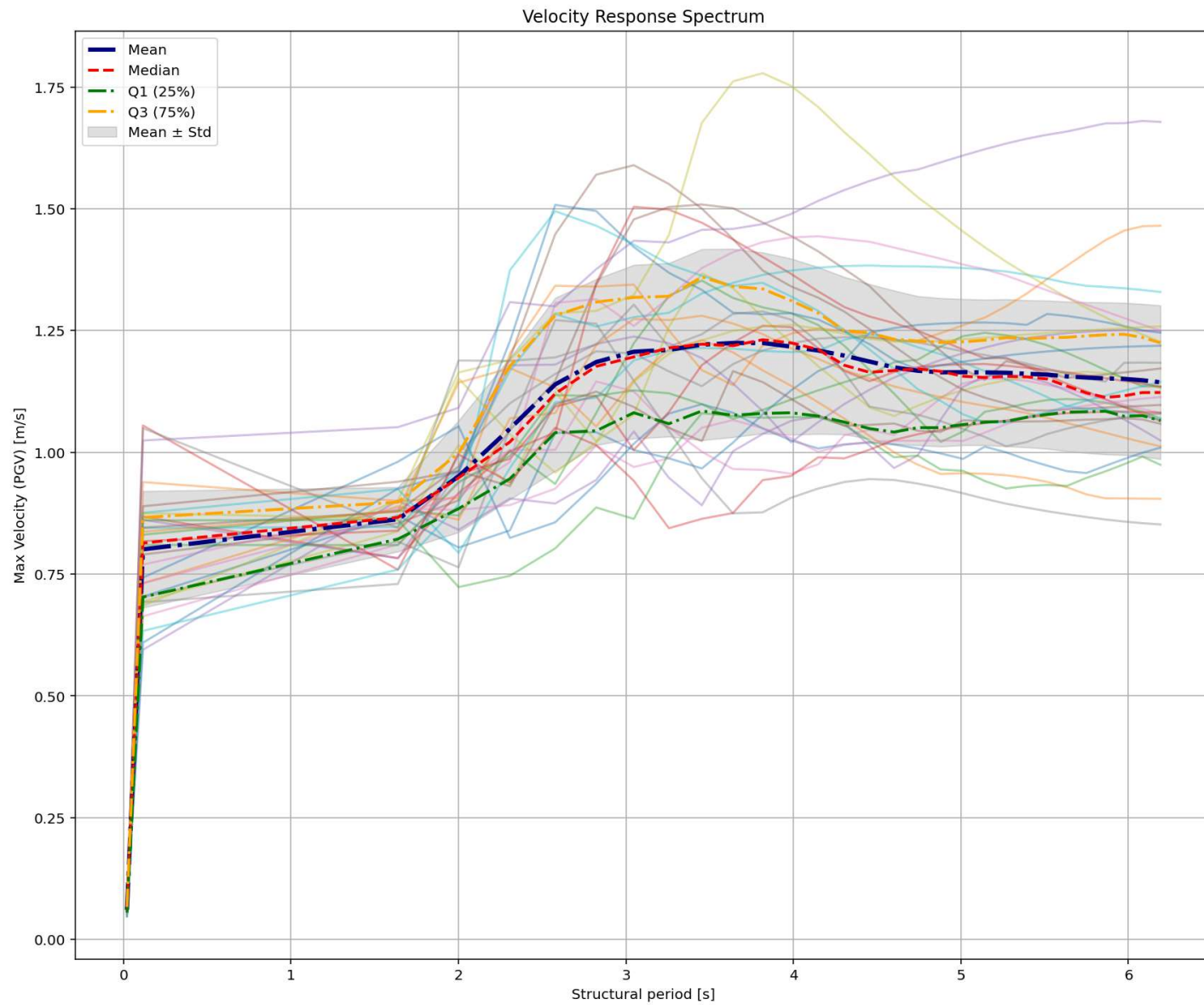
Earthquake Response - Dissipated Energy

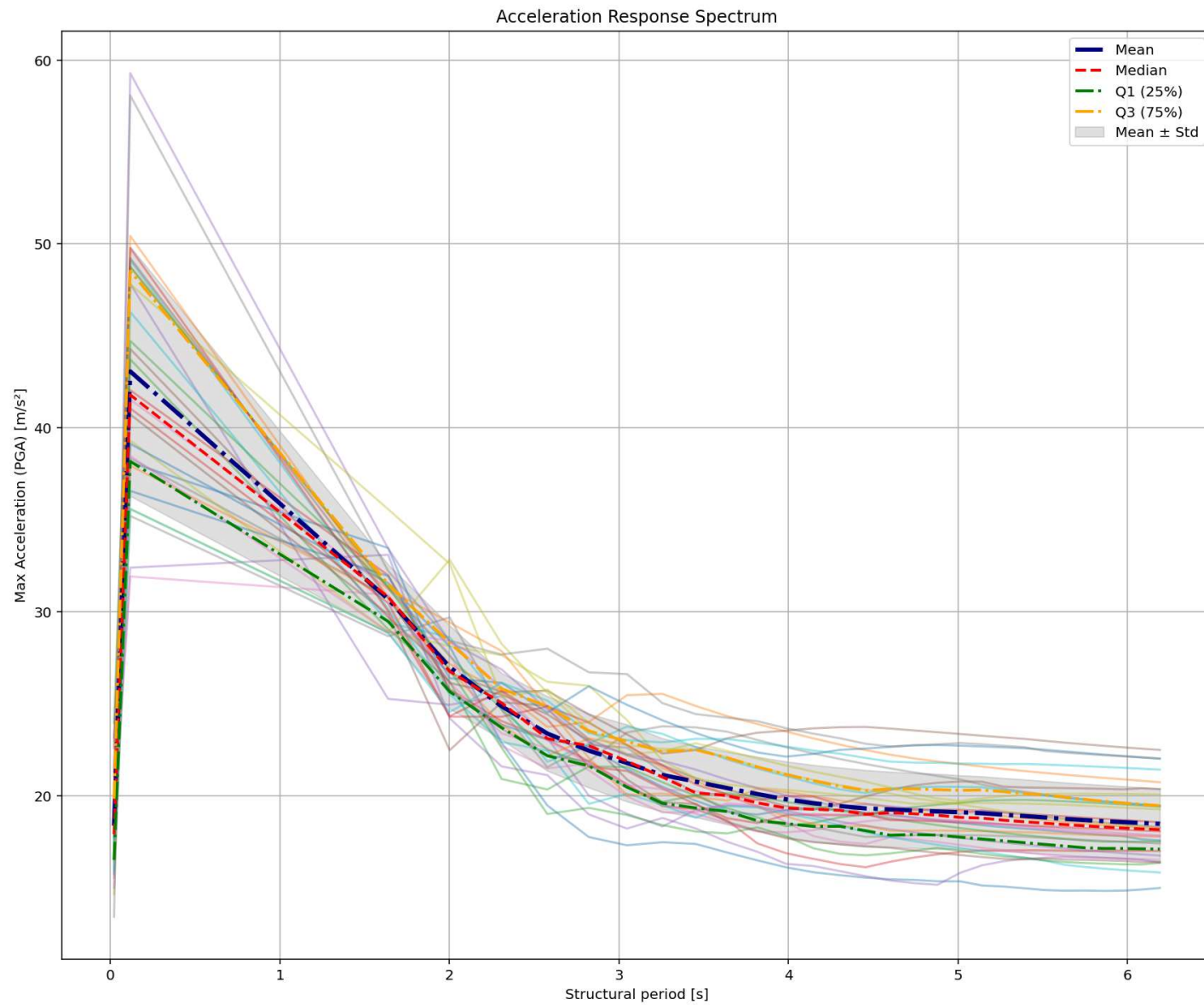


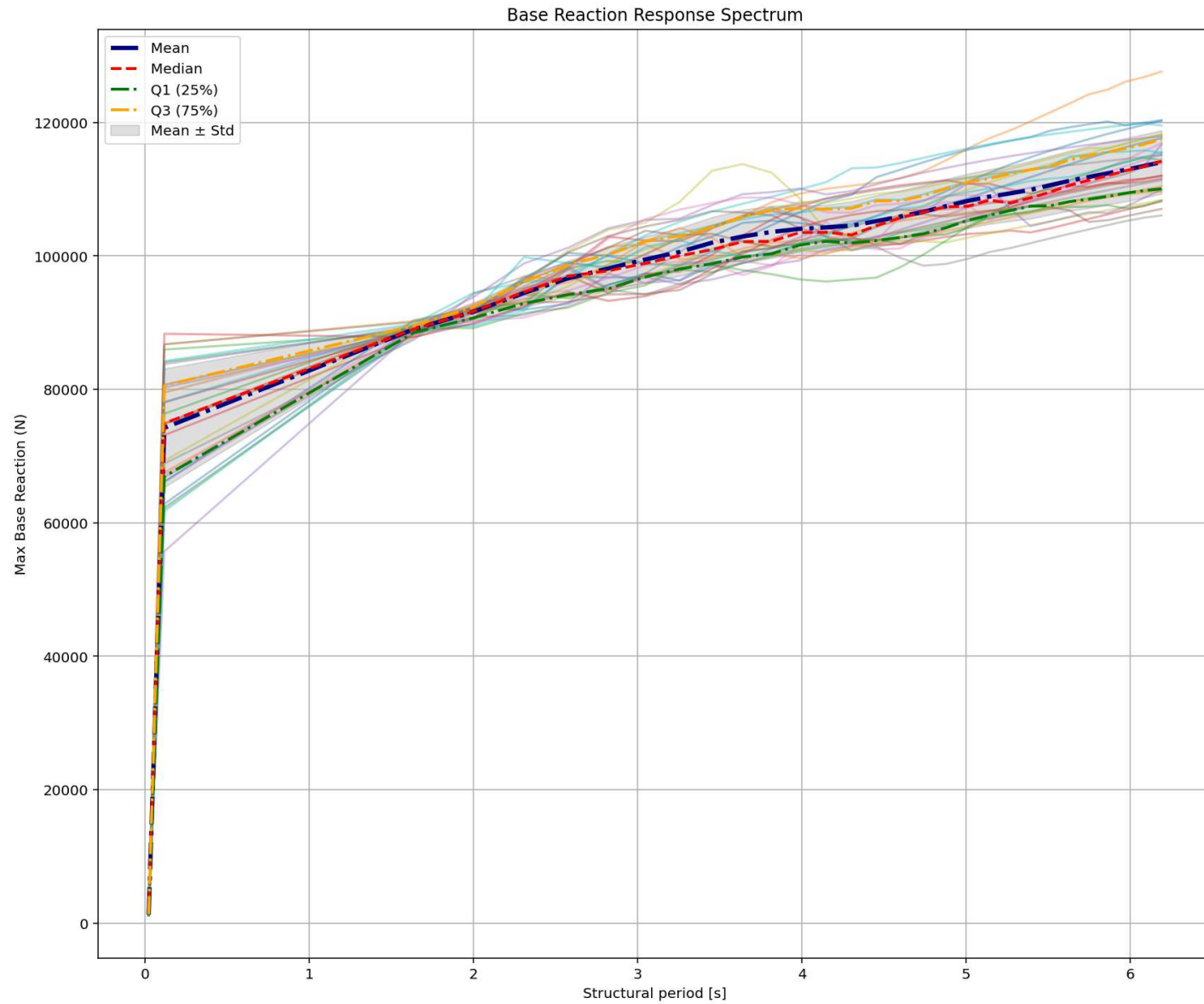
Equivalent viscous damping ratio - Incremental Dynamic Analysis Hysteresis

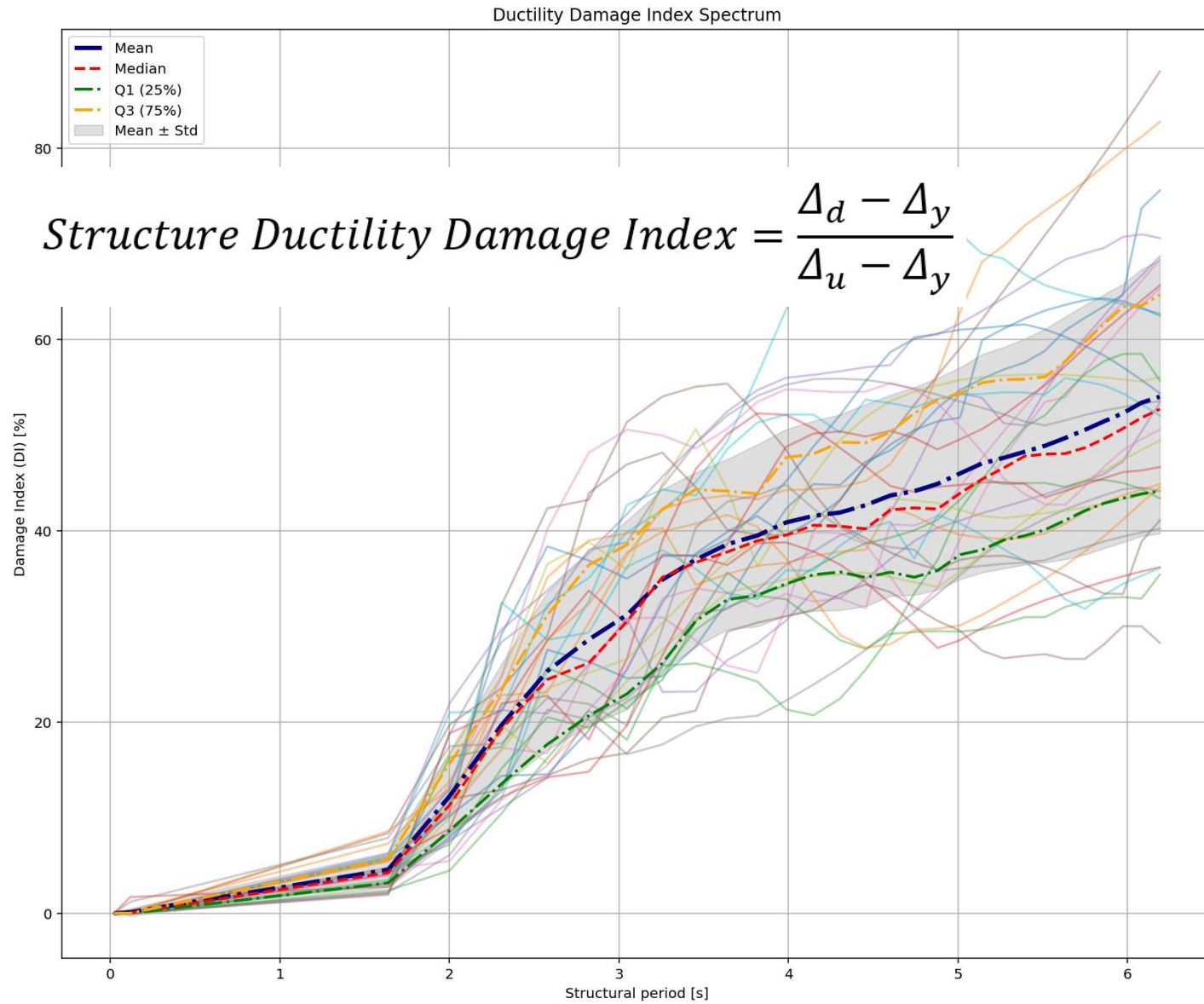


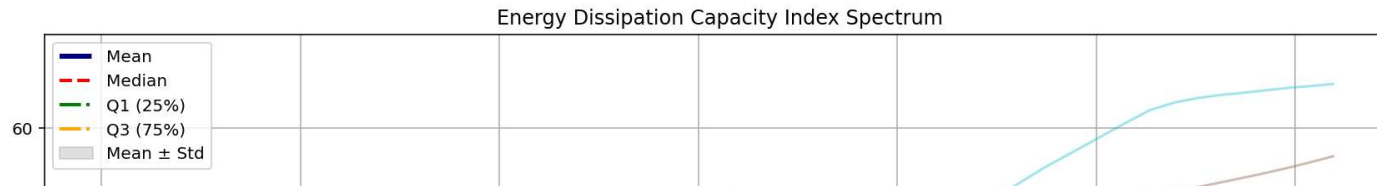




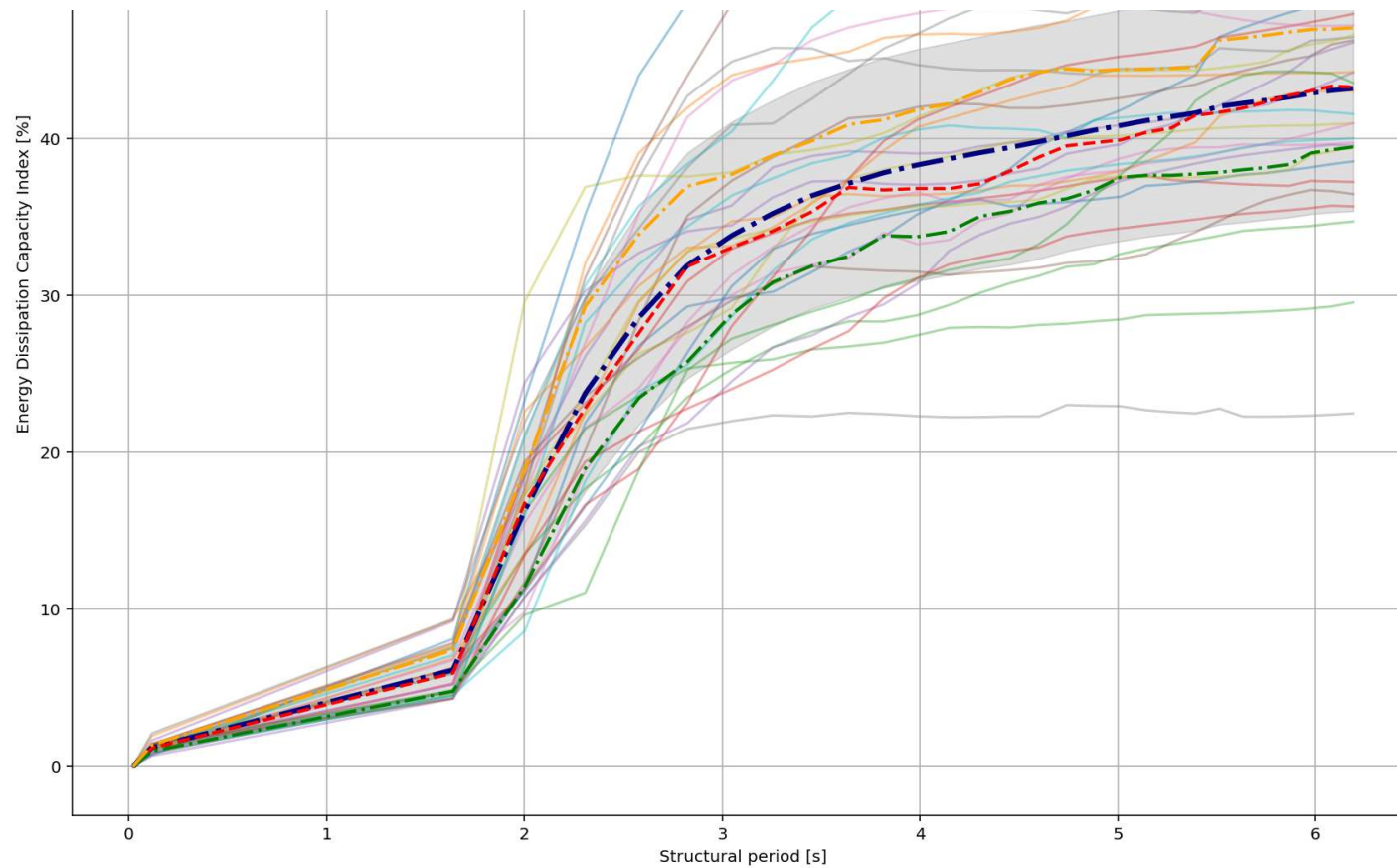


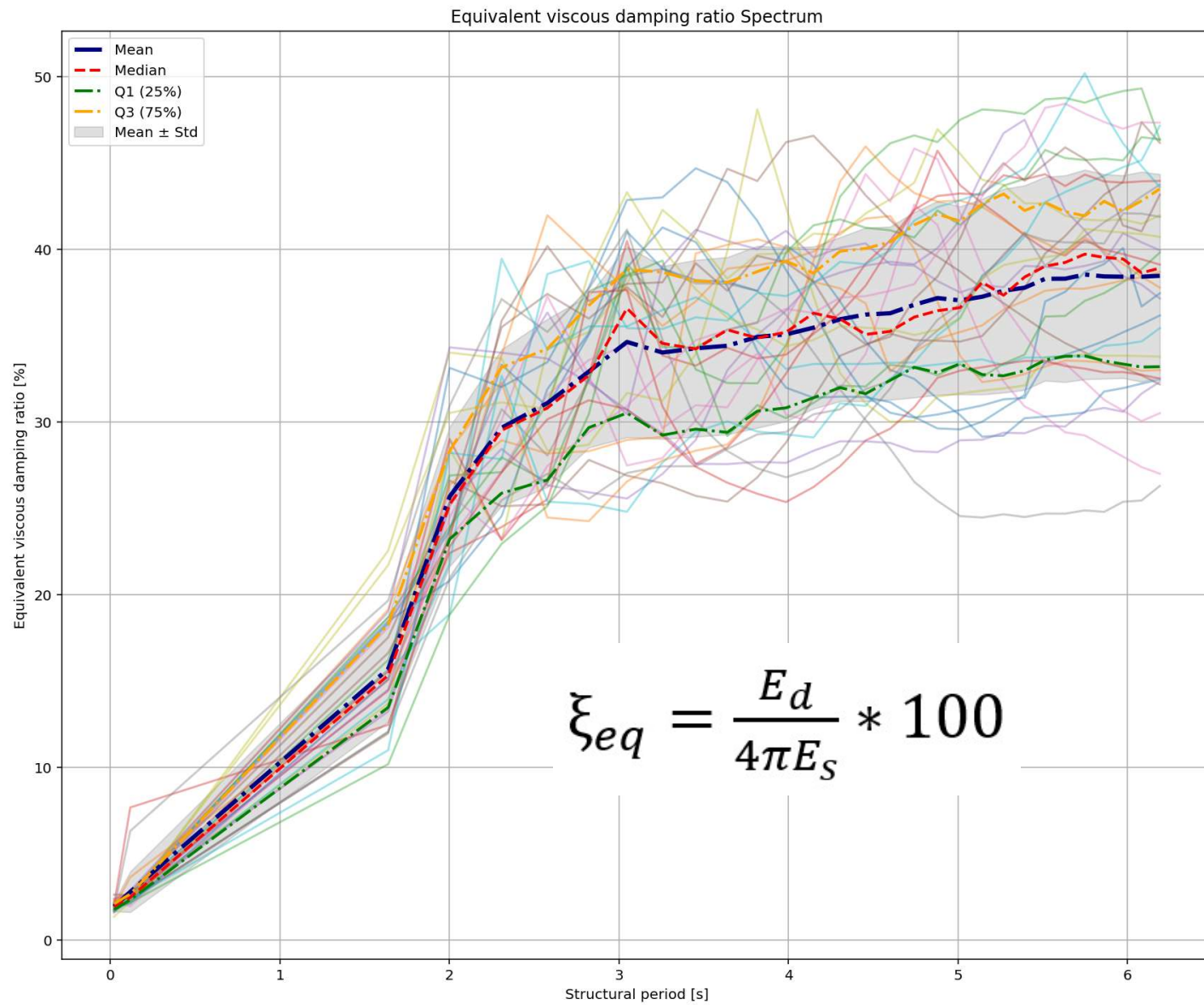




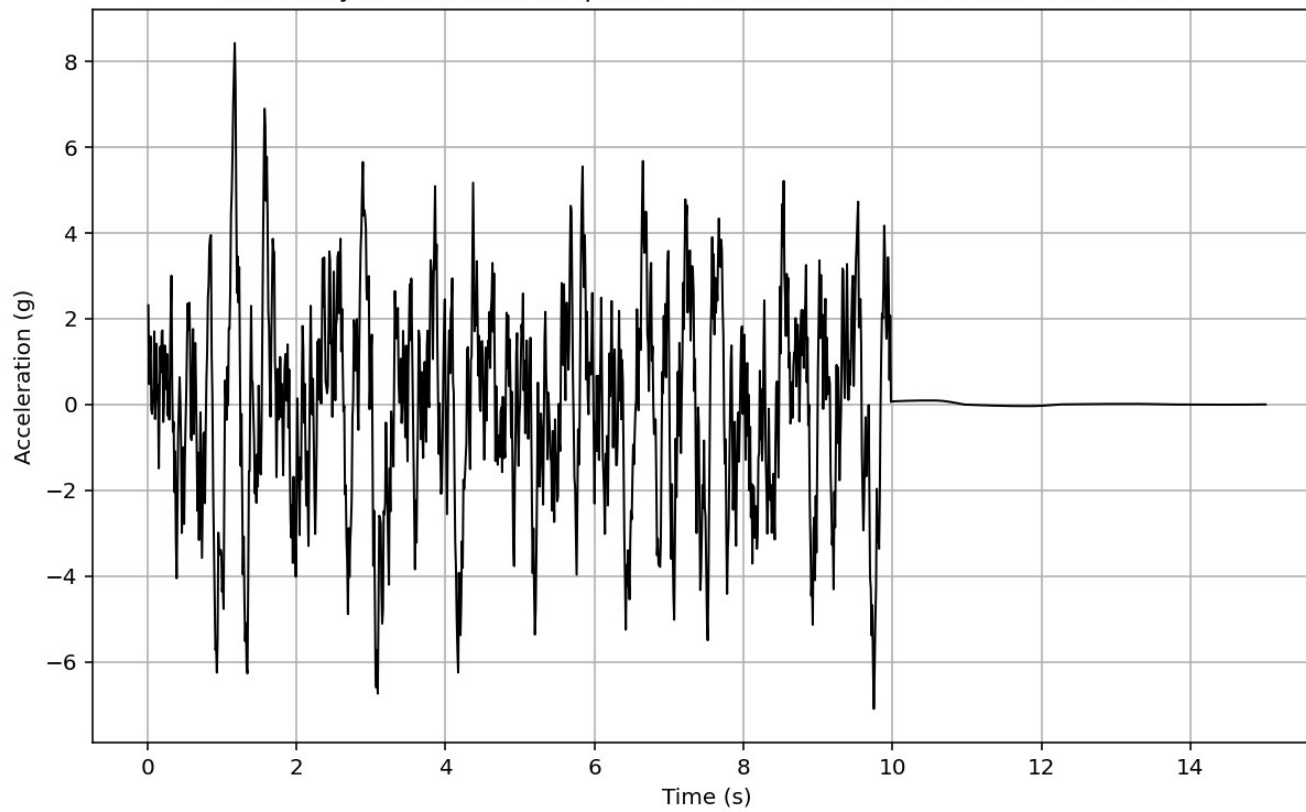


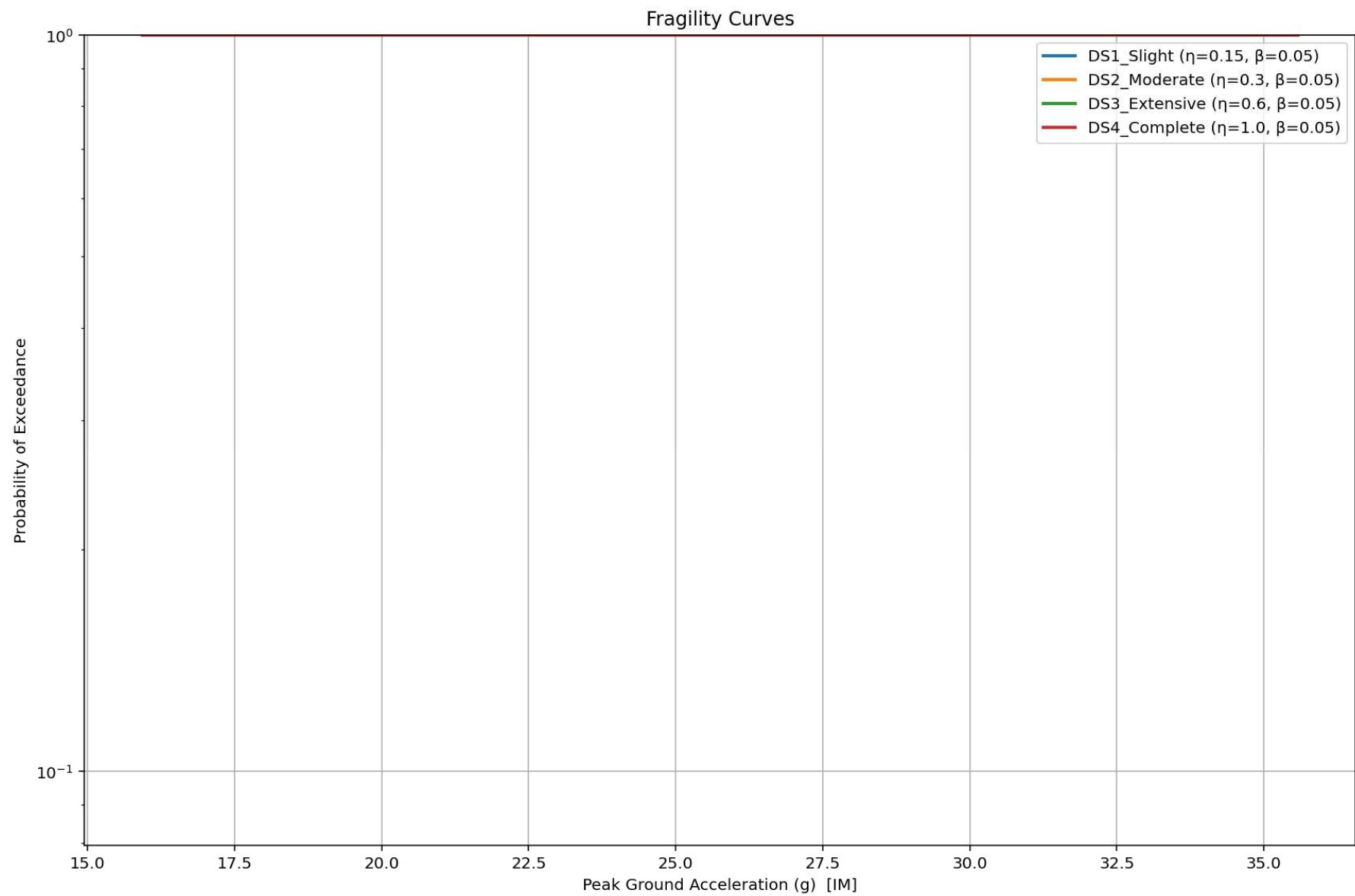
$$\text{Energy Dissipation Capacity Index} = \frac{E_d(\text{Dissipated Energy from Earthquake})}{E_d(\text{Dissipated Energy from Cyclic Displacement})} * 100$$



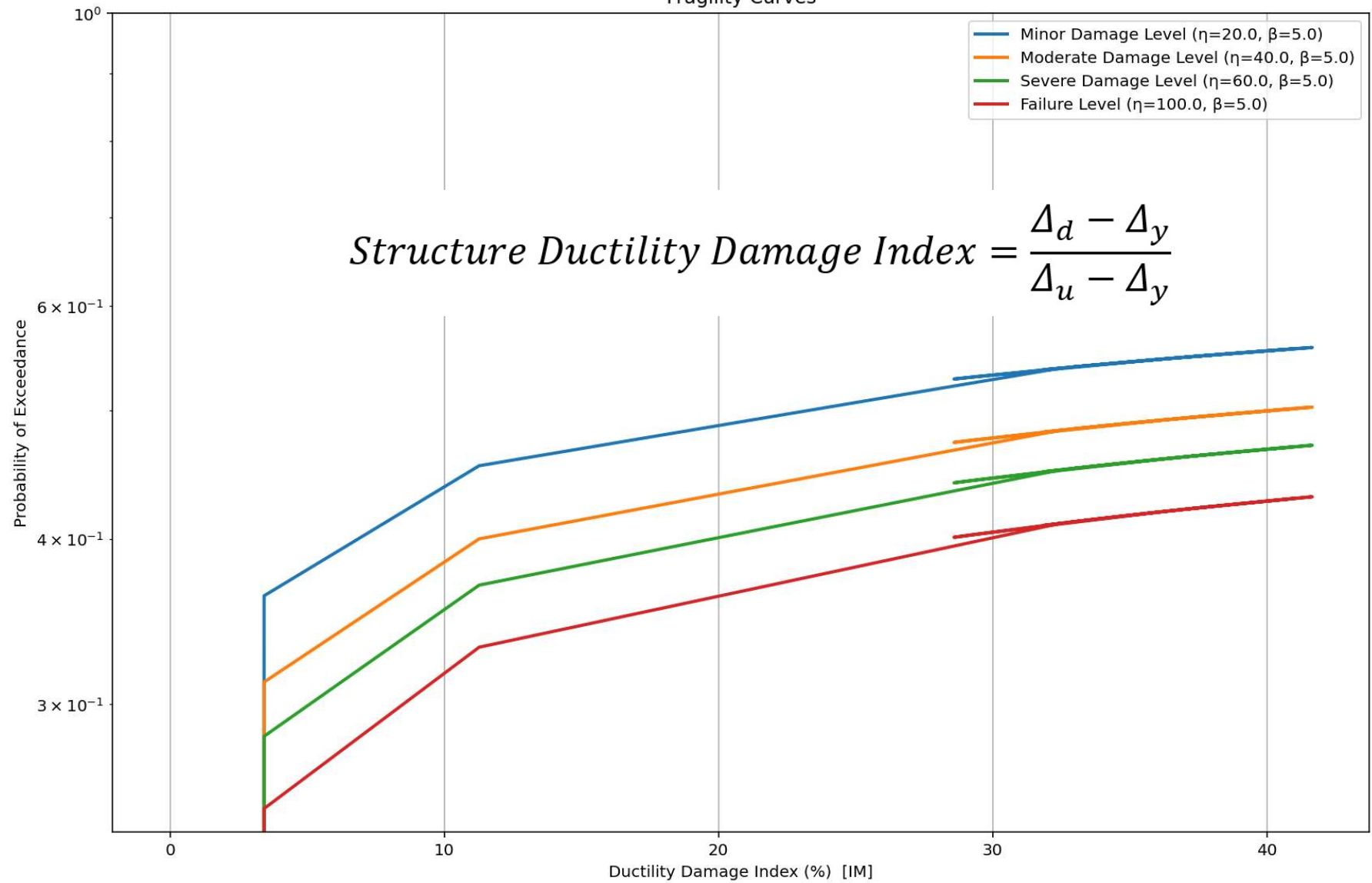


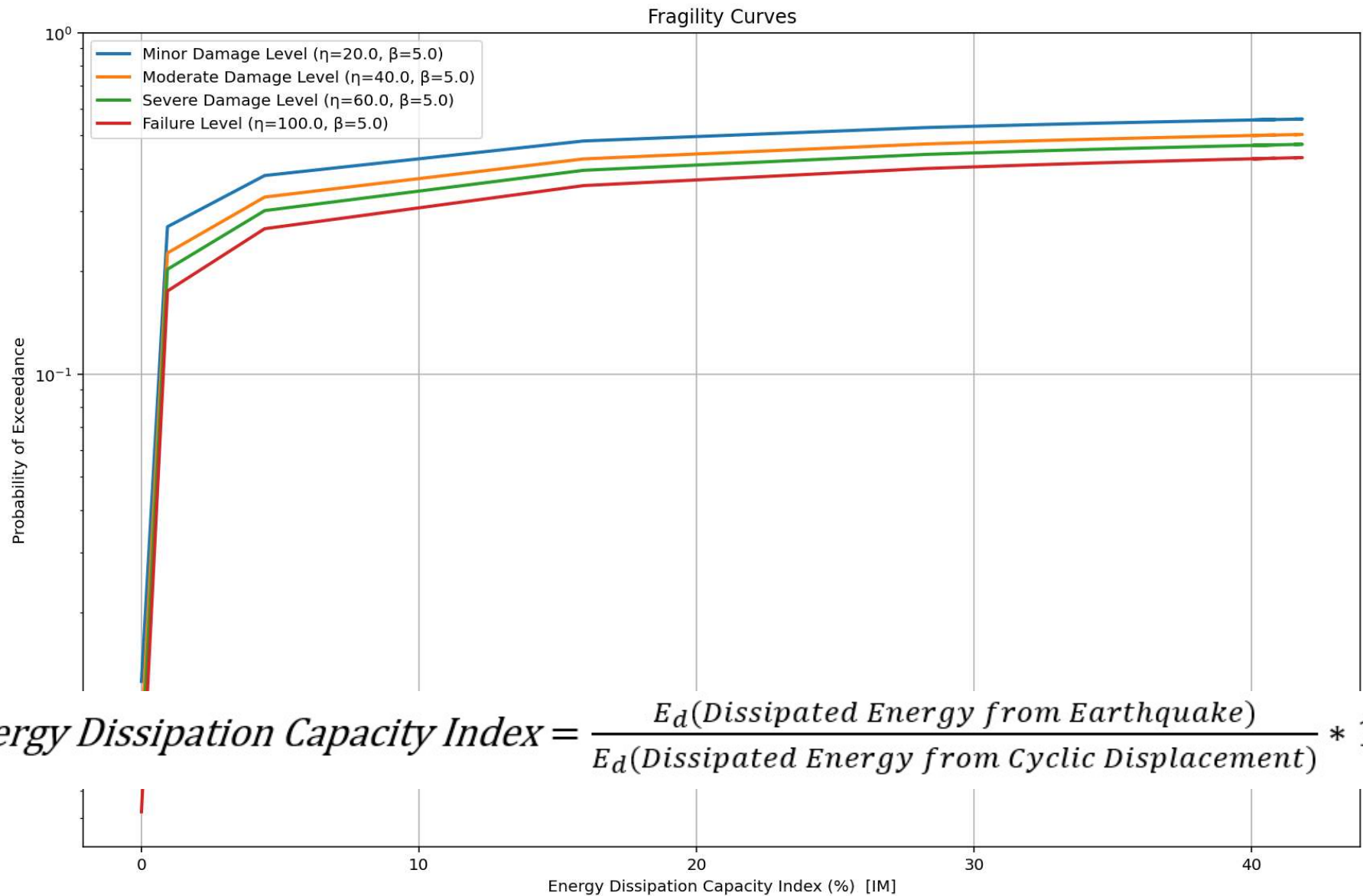
Last Analysis Structural Response + Ground Motion ::: MAX. ABS. : 8.4259



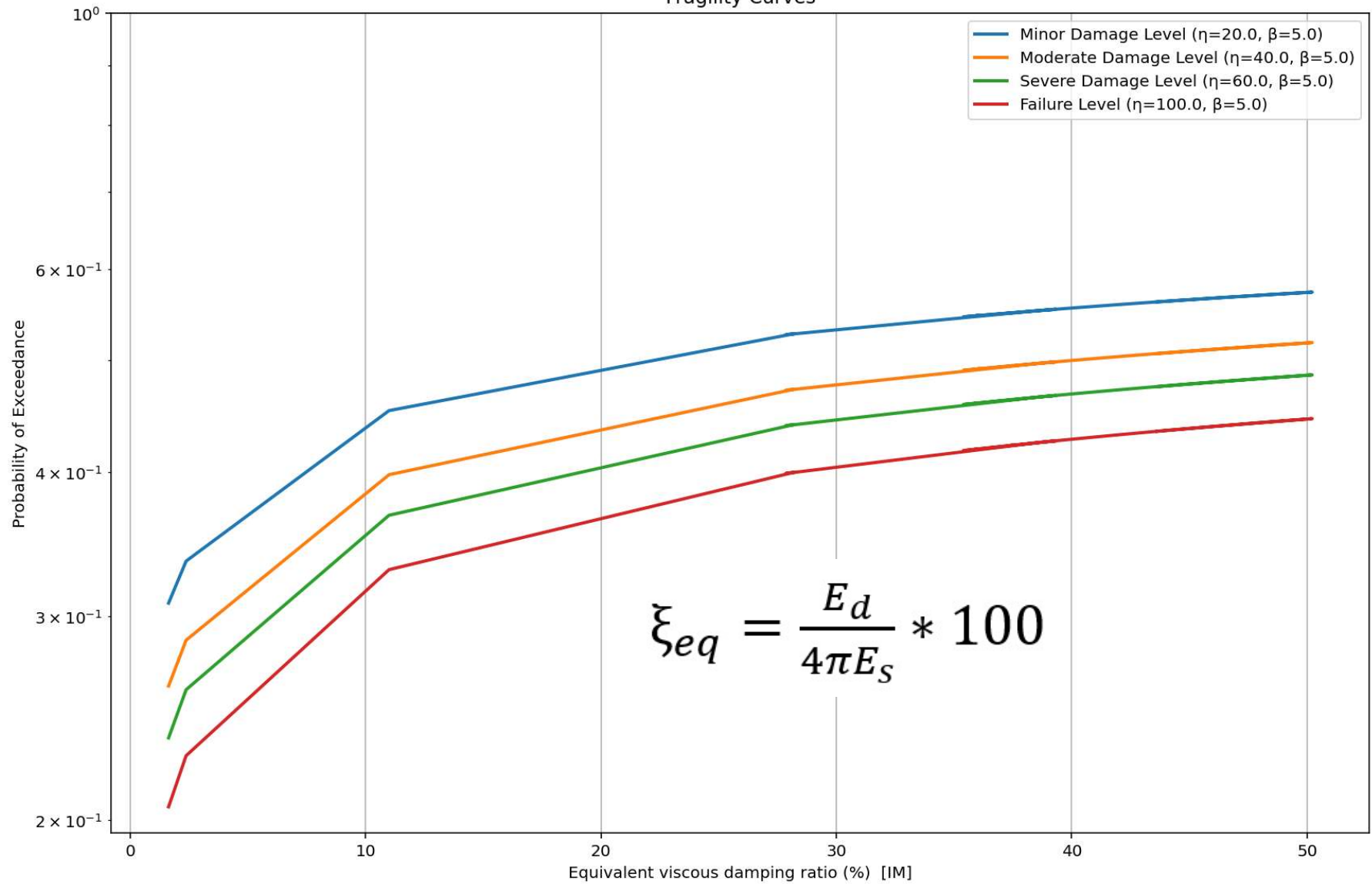


Fragility Curves

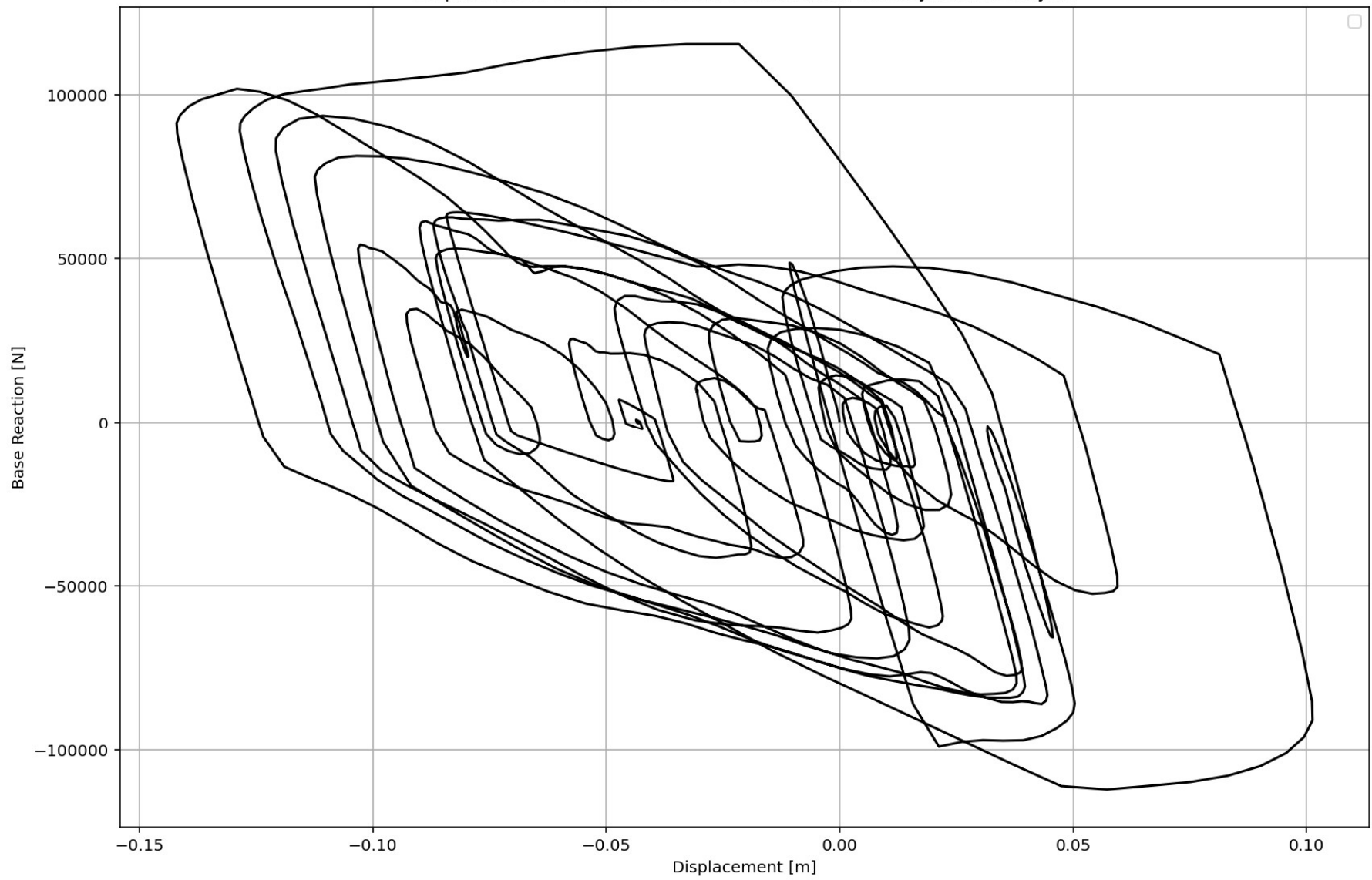


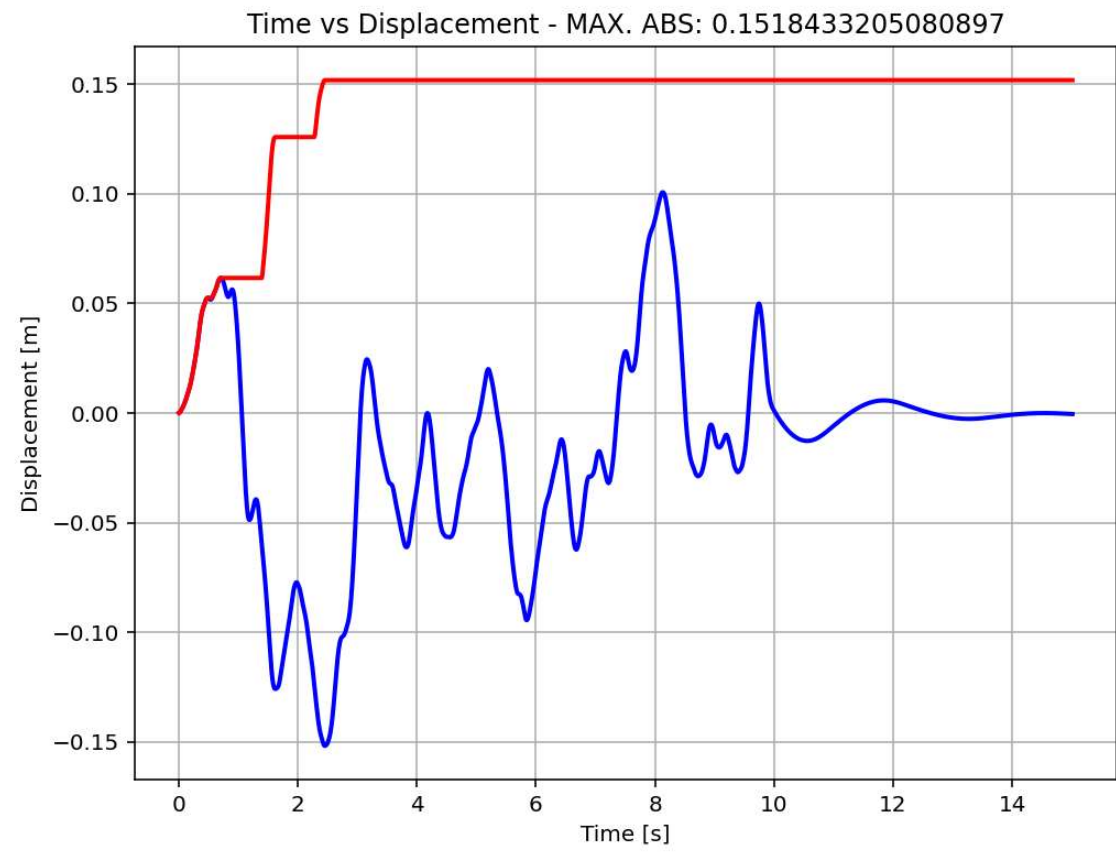


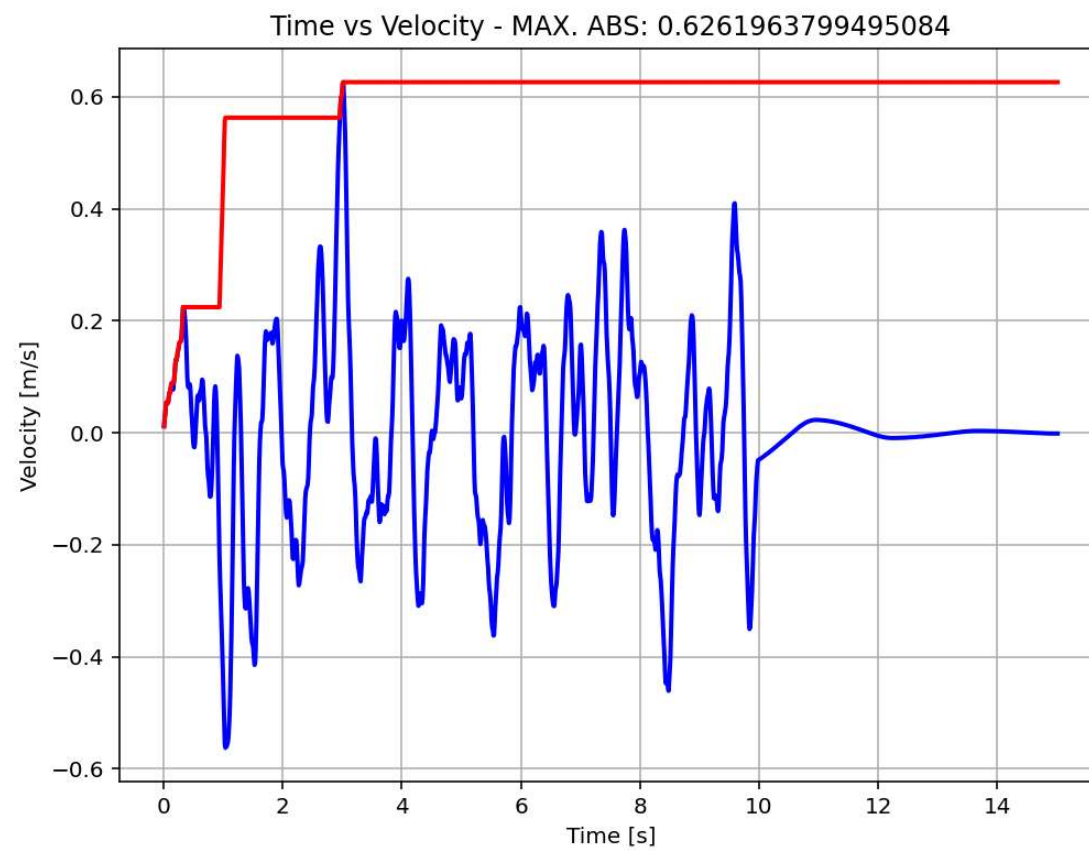
Fragility Curves

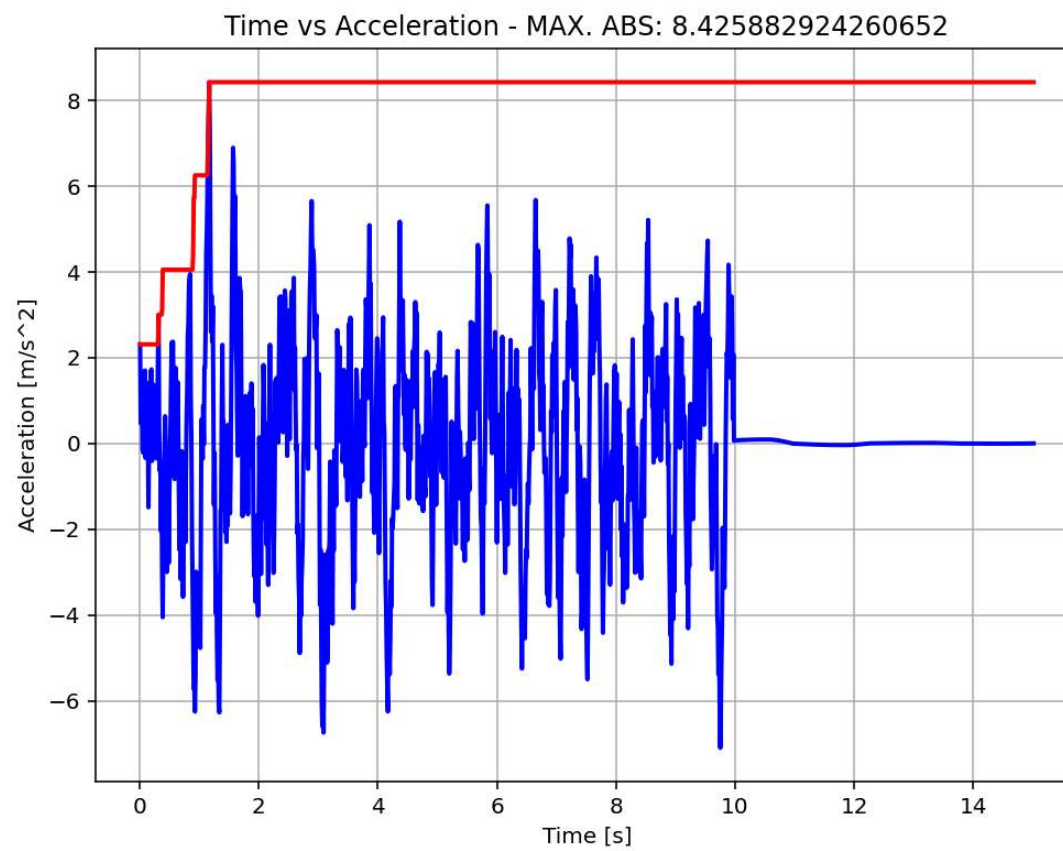


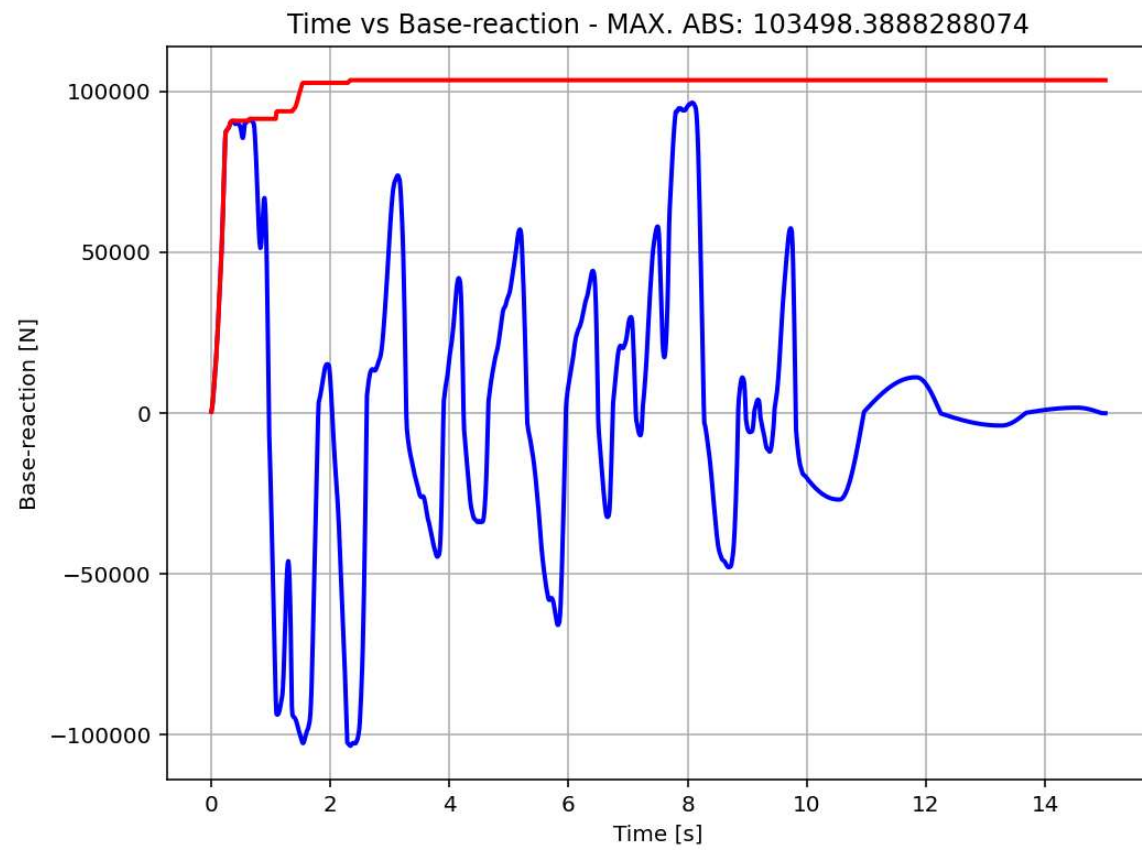
Displacement & Base Reaction Relation From Last Dynamic Analysis

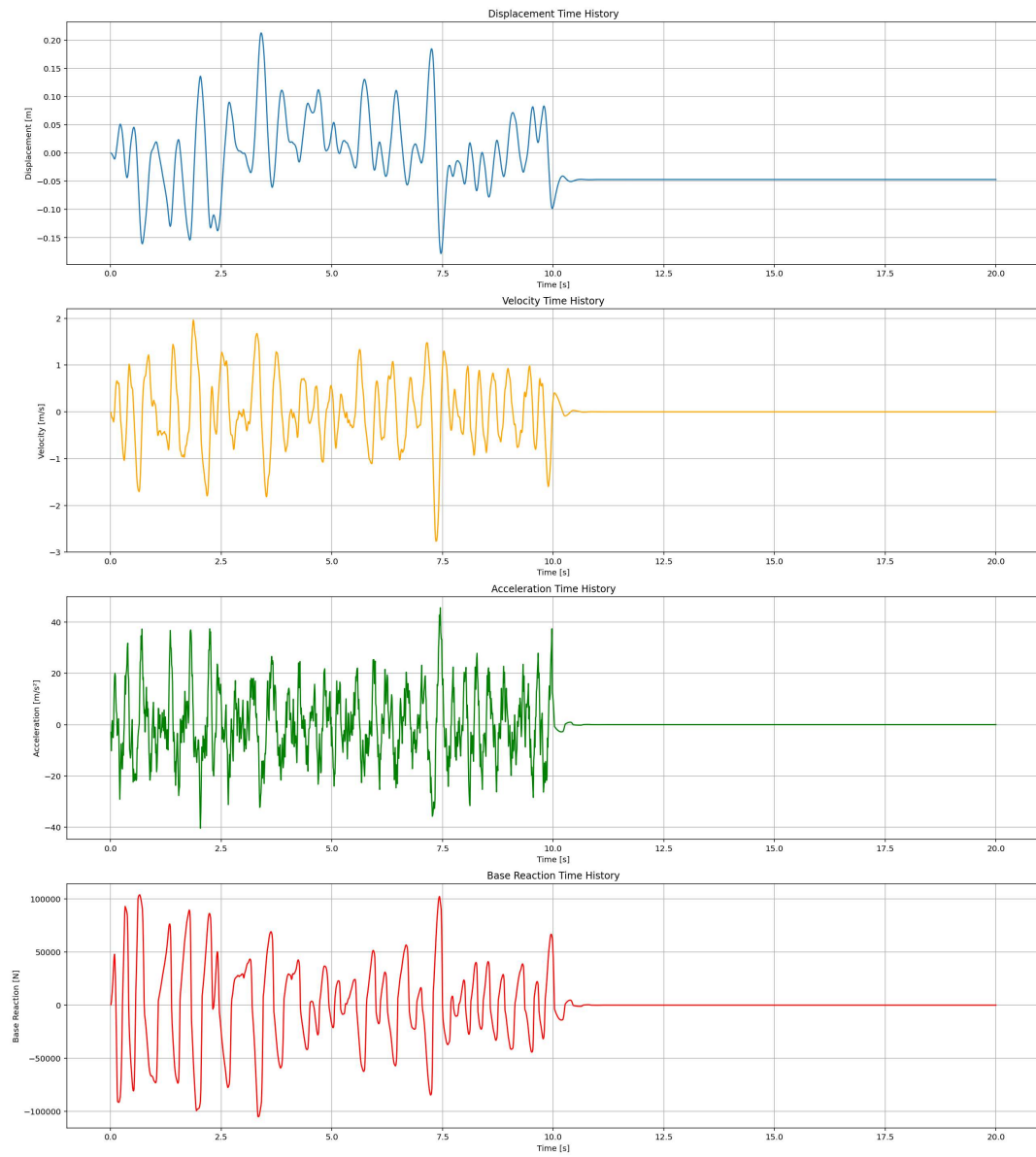












None

